

PROJECT PLAN

Automated Extraction and Interactive Visualization of Wavelength-Domain FOM Data

An eight-week internship project focused exclusively on wavelength-domain FOM data

Duration	8 weeks	Primary user	Research staff working with scientific literature data
Input	Wavelength-FOM Excel; PDF in Phase 2	Output	Wavelength-FOM plots and harmonized Excel files

Executive Recommendation

Recommended scope: combine the two concepts in sequence. During Weeks 1–4, build a reliable Excel-to-plot application. During Weeks 5–8, add a PDF-to-FOM extraction prototype restricted to wavelength-domain FOM rather than attempting to extract every important parameter from every scientific field.

Why this scope is realistic The supplied wavelength-domain FOM spreadsheet can serve as a gold-standard dataset. The intern can compare automated PDF extraction results against this reference and work independently with minimal supervision.

Project Workflow

- 1. Upload structured wavelength-FOM data**
Load the curated wavelength-domain FOM workbook and identify sheets, columns, groups, and missing values.
- 2. Generate interactive plots**
Create editable dot/scatter plots with filtering, hover details, zoom, medians, sample counts, and export options.
- 3. Upload scientific papers**
Accept up to ten PDF files and extract wavelength-domain FOM evidence from text, tables, captions, and figures when readable.
- 4. Validate wavelength-domain FOM records**
Accept wavelength-domain values only, distinguish experiment from simulation, preserve mode/case information, and flag frequency-domain or ambiguous records for exclusion or review.
- 5. Export analysis-ready outputs**
Produce a harmonized wavelength-domain FOM workbook and interactive plots without mixing incompatible domains or origins.

Eight-Week Work Plan

Week	Objective	Main Activities	Required Output
1	Understand the FOM dataset	Inspect the wavelength-domain workbook; document EXP/SIM, grouping, missing-value, and inclusion rules; define the application specification.	Data dictionary and implementation plan
2	Build Excel ingestion	Implement drag-and-drop upload, sheet selection, column detection, numeric conversion, missing-value handling, and data preview.	Working Excel upload module
3	Build interactive visualization	Implement dot/scatter plots, categorical grouping, hover labels, zoom, medians, sample counts, log scale, and configurable axes.	Interactive FOM plotting module
4	Complete Phase 1	Add filters, export functions, error handling, and tests across multiple spreadsheet layouts; prepare a brief user guide.	Stable Excel-to-plot application
5	Start PDF extraction	Implement PDF upload and a structured wavelength-FOM extraction schema; capture value, definition, unit, origin, evidence, and page/figure location.	Single-PDF extraction prototype
6	Apply scientific validation rules	Accept wavelength-domain records only; classify EXP/SIM; preserve mode/case; evaluate canonical FOM definitions; flag frequency-domain, ambiguous, or unsupported records.	Structured wavelength-FOM extraction table
7	Process and evaluate ten PDFs	Run batch extraction on up to ten PDFs; compare results with the supplied wavelength-FOM gold-standard Excel data; calculate precision, recall, and classification accuracy.	Batch prototype and evaluation results
8	Integrate and finalize	Connect PDF extraction, review, harmonization, Excel export, and plotting; fix critical errors; complete documentation and final demonstration.	End-to-end prototype and final report

Phase gate at the end of Week 4 The Excel-to-plot application must be independently usable before PDF extraction is attempted. This guarantees a complete deliverable even if PDF extraction requires additional refinement.

Functional Scope

Phase 1 — Excel-to-Plot Application (Weeks 1–4)

Capability	Minimum Requirement
Input	Drag-and-drop .xlsx and .csv files; support workbooks with multiple sheets.
Data handling	Detect numeric and categorical columns; identify missing or malformed values without modifying the source file.
Domain control	Include wavelength-domain FOM records only; exclude or flag frequency-domain and ambiguous records.
Origin control	Allow EXP and SIM records to be filtered or displayed separately.
Visualization	Generate interactive dot/scatter plots with hover details, zoom, configurable groups, medians, sample counts, and optional log scale.
Output	Export plots and the filtered analysis dataset.

Phase 2 — PDF-to-Wavelength-FOM Extraction (Weeks 5–8)

The extraction module should target wavelength-domain FOM evidence only. Frequency-domain FOM and general extraction of all important values from arbitrary scientific papers are outside the eight-week scope.

Field	Example	Validation Requirement
FOM reported	Yes / No / Unclear	Do not infer that FOM was reported without explicit evidence.
FOM value	2.1	Preserve the reported value and associated mode or case.
Definition	FOM = S / FWHM	Store the author's definition or mark it as not stated.
Domain	Wavelength / Other / Unclear	Only wavelength-domain FOM records may enter the final output.
Origin	EXP / SIM / UNCLEAR	Keep experimental and simulated results identifiable.
Sensitivity and FWHM	642 nm/RIU; 3 nm	Check that sensitivity and linewidth are both wavelength-domain quantities.
Mode / case	Mode 1; final design	Create separate records when multiple modes or cases are reported.
Evidence	Short supporting text	Every accepted record requires traceable evidence.
Location	Page 6; Fig. 4	Record page, figure, or table location when available.
Review status	Approve / Edit / Exclude	No uncertain AI-generated record should be silently finalized.

Evaluation, Deliverables, and Supervision

Evaluation Framework

Metric	Question Answered	Expected Reporting
Precision	How many extracted FOM records are correct?	Report overall and by field.
Recall	How many gold-standard FOM records were found?	Report missed records and causes.
Domain validation	Were wavelength records accepted and frequency/ambiguous records excluded?	Report acceptance and exclusion errors.
Origin accuracy	Were EXP, SIM, and UNCLEAR distinguished correctly?	Report confusion counts.
Evidence accuracy	Does each value point to the correct supporting text or location?	Report verified percentage.
Time reduction	Does the application reduce manual extraction and plotting time?	Compare a small timed sample.

Final Deliverables

No.	Deliverable
1	Working Excel-to-plot application
2	PDF-to-wavelength-FOM extraction prototype supporting up to ten PDFs
3	Source code with setup instructions
4	Raw extraction and harmonized Excel outputs
5	Evaluation report using the supplied FOM ground-truth data
6	Short user guide and final demonstration slides

Required Inputs from the Supervisor

Input	Purpose
Wavelength-domain FOM Excel file	Gold-standard reference data and Phase 1 input
Ten corresponding PDF papers	Development and evaluation of Phase 2
One-page classification rules	Domain, origin, canonical FOM, and inclusion logic
One preferred plot example	Visual target for the plotting interface

Minimal Supervision Plan

Checkpoint	Supervisor's Role	Estimated Time
Project start	Provide files, rules, and expected output example.	About 60 minutes
End of Week 4	Confirm that the Excel-to-plot application is usable.	About 20 minutes
End of Week 8	Review the final demonstration and evaluation report.	About 30 minutes

Final scope decision A simple Excel-to-scatter-plot tool is achievable but too limited for an eight-week internship. A universal paper-mining system is too ambitious. The recommended middle scope—Excel visualization plus PDF extraction restricted to wavelength-domain FOM—offers the best balance of feasibility, independence, and research value.